

Write Strength, Not Write Count: The β Range Dominates Householder Arity in Gated Delta-Rule Attention

Eric Wu^{1,2}

¹Stanford University

²Alpha AI Engineering

Abstract

Delta-rule linear attention with a channel-wise forget gate can be made more expressive in two ways: by applying more Householder factors per token (*arity*, R), or by widening the range of the write-strength gate β from the strict $(0, 1)$ to the reflection range $(0, 2)$. Recent work has concentrated on the first. We report a controlled factorial measurement suggesting the second matters far more. Our design crosses $R \in \{1, 2\}$ with β regime over two non-solvable group word problems (A_5 , S_5) and twelve seed bundles, and adds three further factors so that the confounds such a comparison invites are removed by construction rather than argued away: a *capacity* factor giving each $R=1$ arm a parameter-matched twin (-0.055%) that carries the $R=2$ parameter count in feed-forward width, in *both* regimes; a *learning-rate* factor spanning $33\times$ on the arms where a gap exists; and an *arity ladder* extending strict β to $R \in \{3, 4\}$ with a capacity control at each rung, reaching the regime where the $k \leq 2$ bound of Grazi et al. [?] no longer applies. 216 cells, 16.28 GPU-hours, all fully paired at the seed-bundle level. The β widening at fixed R is large in every position band where both arms are above chance, while doubling R under strict β is consistently positive but one to two orders of magnitude smaller — and the β widening is *free*, adding zero parameters, whereas doubling R costs $+40.3\%$. The interaction is not a capacity artifact: matching moves the A_5 mid-range estimate *up*, to $+58.88\text{pp}$. Nor is it an optimization artifact: no rate closes the strict arms’ deficit at a length *inside* the training range, where they reach stable near-zero training loss and still fall 9–21pp short — a generalization gap, not a training failure. At $n = 12$, **four** capacity-matched bands survive one-sided Holm correction, the strongest at $p = 0.00003$. Because Kimi K3’s KDA layer fixes $\beta = \text{Sigmoid}(\cdot) \in (0, 1)$, a determinant argument shows K3-style nodes cannot obtain a negative-determinant transition at any R ; whether they would benefit from $R > 1$ in practice is a separate question our scale cannot settle. We report results against our own interest. Our pre-registered mechanism — that reflections help because S_5 ’s odd generator demands $\det = -1$ — is *contradicted*, and the fallback explanation we offer is contradicted too. The arity ladder is the sharper case: its one positive cell is capacity-matched, and decomposing that match shows 81% of the $R=4$ “gain” is damage the control’s feed-forward block does to itself, leaving a raw arity effect of $+3.11\text{pp}$ at $t = 1.06$. We report the effects, their sizes, and the reasons each is suggestive or established.

1 Introduction

Linear-attention sequence models trade the quadratic cost of softmax attention for a fixed-size recurrent state, and the delta rule [6, 8] has become the dominant way to write into that state: each token performs a rank-one *erase* followed by a rank-one *write*. A well-documented consequence is limited state-tracking ability. Two remedies have been proposed. DELTAPRODUCT [7] increases the number of Householder factors applied per token, a quantity we call the *arity* R . Separately, Grazi et al. [3] showed that admitting *negative* eigenvalues in the state transition unlocks state tracking that is unreachable otherwise; in a delta-rule node this is achieved by widening the write-strength gate β from the strict range $(0, 1)$ to the reflection range $(0, 2)$, since $1 - \beta$ is the non-unit eigenvalue of each factor.

These two levers are not independent, and the literature has largely pursued the first. That matters practically, be-

cause arity is expensive — each extra factor needs its own key and gate projections — while widening β is a one-character change to a sigmoid that adds no parameters at all. It also matters now, because the largest deployed delta-rule attention variant, the Kimi Delta Attention (KDA) layer of Kimi K3 [5], specifies $\beta_t = \text{Sigmoid}(W_\beta x_t) \in (0, 1)$ — strictly inside the non-reflective range.

We test this with a fully paired factorial design at otherwise identical geometry, on two non-solvable group word problems. Its core crosses arity $R \in \{1, 2\}$ with β regime over twelve seed bundles. Because that core alone cannot distinguish arity from the parameters arity buys, nor expressivity from trainability, the design carries three further factors aimed at exactly those alternatives: a *capacity* factor pairing each $R=1$ arm with a parameter-matched twin in both regimes, a *learning-rate* factor spanning $33\times$, and an *arity ladder* carrying strict β to $R \in \{3, 4\}$ with a matched control at each rung. 216 cells in total. The intent is that the objec-

tions a reader would raise first are answered by the design rather than by discussion.

The design is deliberately narrow: it is built to estimate one interaction term rather than to establish a state of the art. It is also small, and we have tried to be scrupulous about what a 1.4M-parameter probe can and cannot support. Two things in particular govern how our results should be read, and we put them here rather than in a late limitations section.

The metric is a prefix average, and we analyze around it. Our harness scores mean token accuracy over *all* positions $1..L$ of each evaluation sequence. Consequently the five nominal “lengths” are nested, and a model that is correct below $L = 40$ and at chance above it still produces a smooth, gracefully-decaying curve that looks like partial extrapolation. For our strict- β arms this is exactly what happens: a zero-parameter model asserting “correct below 40, at chance above” reproduces their entire length curve to within 1.3pp (Table 3). We therefore report *de-nested position bands* as our primary analysis (Section 5.1) and treat the prefix-averaged numbers as secondary. De-nesting strengthens our A_5 result and *eliminates* two of the seven apparently significant S_5 cells, and we report both directions.

Both strict- β arms are at chance beyond $L = 40$. In every de-nested band past position 40, R1 and DP2-strict score within $2\times$ chance (Table 2). This has a consequence we must state plainly, because it defeats an argument we would otherwise have liked to make: the strict-regime arity contrast cannot bound what capacity is worth, since it is measured between two models that are both floored. This is precisely why the capacity factor is part of the design rather than a bound computed after the fact (Section 5.4).

Contributions. (i) A controlled factorial measurement of arity against β range at matched geometry, fully paired at the seed-bundle level so every arm within a bundle is scored on a byte-identical evaluation bank, with capacity and learning-rate factors built in so that the two leading alternative explanations are addressed by construction (Section 4). (ii) The finding that in every band where both arms are measurable, the β effect exceeds the arity effect by one to two orders of magnitude at zero parameter cost (Section 5.3), together with a positive $\beta \times R$ interaction in five of eight de-nested bands (Section 5.1) that survives capacity matching and is not closed by any learning rate we tried (Sections 5.4–5.5). (iii) A determinant observation about K3-style nodes, and a correction to a tempting but wrong reading of K3’s new bounded-decay parameterization (Section 3). (iv) Two negative results against our own hypotheses: the group-parity account of *why* reflections help is contradicted, and so is the headroom account we offered as a replacement (Section 6). (v) A characterization of the strict- β deficit as an *in-distribution generalization gap* rather than a training failure — these models fit to near-zero loss and still miss inside the training range (Section 5.5) — and a free replication check quantifying what a single cell of

this apparatus can resolve (Section 5.7).

We claim no new operator and no new theory. The gated Householder operator we measure is already implemented in public kernels and published at scale [2]; the determinant argument of Section 2.3 is a special case of a published result [3], and weaker than it. What is new is the controlled measurement. This is an empirical, partly negative result and we frame it as one.

2 Background

2.1 The gated delta rule

Let $S_t \in \mathbb{R}^{d_k \times d_v}$ be the recurrent state, k_t the (L2-normalized) key, v_t the value, β_t the write strength and $\alpha_t \in (0, 1)^{d_k}$ a channel-wise forget gate. The gated delta rule is

$$S_t = (I - \beta_t k_t k_t^\top) \text{Diag}(\alpha_t) S_{t-1} + \beta_t k_t v_t^\top, \quad (1)$$

with readout $\tilde{o}_t = S_t^\top q_t$. Equation (1) is exactly the KDA recurrence of Kimi K3, and specializes to gated DeltaNet [9] and, with α scalar per head, to Mamba-2-style gating [1].

2.2 Householder arity

DELTAPRODUCT [7] applies R rank-one factors per token, each with its own key and write strength:

$$S_t = \left[\prod_{i=1}^R (I - \beta_t^{(i)} k_t^{(i)} k_t^{(i)\top}) \right] \text{Diag}(\alpha_t) S_{t-1} + \text{writes}. \quad (2)$$

Each factor is a true Householder reflection when $\beta = 2$.

Notation. We write R for the number of factors per token and call it *arity*. DELTAPRODUCT writes n_h , denoting models $\text{DELTAPRODUCT}_{n_h}[\cdot]$ with the admissible eigenvalue range in brackets, and does not use the word “arity”; our R is their n_h , and our arms are their $n_h \in \{1, 2\}$ (their $n_h=1$ is DeltaNet). Arity is not free: the $R=2$ node needs a second set of key and gate projections.

2.3 The determinant fact, and to whom it belongs

The per-token map acting on S_{t-1} in Equation (2) is $A_t = \left[\prod_i (I - \beta_t^{(i)} k_t^{(i)} k_t^{(i)\top}) \right] \text{Diag}(\alpha_t)$. For a unit-norm key, $\det(I - \beta k k^\top) = 1 - \beta$, so

$$\det A_t = \left[\prod_{i=1}^R (1 - \beta_t^{(i)}) \right] \cdot \prod_j \alpha_{t,j}. \quad (3)$$

Since $\alpha_{t,j} > 0$ by construction, the *sign* of $\det A_t$ is governed entirely by the β factors, and if every $\beta^{(i)} \in (0, 1)$ then $\det A_t > 0$ for every R : raising arity cannot recover a negative determinant that the β range has excluded.

What this does *not* predict. We are careful here, because Equation (3) does not by itself predict the interaction we go on to measure. At $R = 2$ with $\beta \in (0, 2)$, two negative factors multiply to a *positive* determinant, so $R=2$ reflection unlocks no determinant sign that $R=1$ reflection cannot already reach. Read strictly, Equation (3) predicts a main effect of β and *no* $\beta \times R$ interaction at $R = 2$. Our measured interaction is therefore not explained by this argument, and we do not present

it as confirmed theory. The determinant argument licenses exactly one claim — that strict β forecloses negative determinants at every arity — and we restrict ourselves to that.

Attribution. Grazzi et al. [3] study $\mathcal{M}_k^n(\Omega)$, the products of exactly k generalized Householder matrices $I - \beta_i v_i v_i^\top$ with $(1 - \beta_i) \in \Omega$, $\|v_i\| = 1$. Their Proposition 1.3 states that any eigenvalue of $N \in \mathcal{M}_k^n((-1, 1])$ is either 1 or has modulus below 1, and that if in addition $N \in \mathcal{M}_k^n([0, 1])$ and $k \leq 2$, then it lies in $[0, 1] \subset \mathbb{R}$. The $k \leq 2$ hypothesis is essential — it was added in a published correction, and its failure for $k \geq 3$ is what DELTAPRODUCT exploits. They conclude such transitions are the identity or non-orthogonal, hence neither reflections nor rotations, and that for $k \leq 2$ the model cannot learn parity by their Theorem 1. Their Proposition 1.2 is the permutation statement proper: realizing a permutation matrix requires the eigenvalue set $\{-1, 1\}$. Equation (3) is a determinant-level restatement, for the delta rule with normalized keys, of a spectral fact they establish in far greater generality; it also says strictly less, since at $R \geq 3$ a positive determinant no longer implies non-negative real eigenvalues. We claim no theoretical novelty for it whatsoever.

3 The K3 KDA node

Kimi K3 [5] is a mixture-of-experts model with 2.78T total and 104.2B activated parameters, a 1M-token context, and native vision. Each block places three KDA layers followed by one gated MLA layer — a 3:1 ratio *per block* (§2.1) — with one additional MLA layer at the end of the backbone, giving 69 KDA and 24 MLA layers (Table 1, §3.2) and a global ratio of 2.875:1. K3 uses no explicit positional encoding (§3.4) and applies NoPE to all MLA layers, so positional and recency information reaches the model only through KDA’s gating and decay (§2.1.2). The KDA recurrence is Equation (1), and its write strength is

$$\beta_t^h = \text{Sigmoid}(W_\beta^h x_t) \in (0, 1), \quad (4)$$

a plain sigmoid, per head and per token, with no scale factor: we searched the report and β is never scaled, doubled, or rescaled anywhere, with the open interval stated in two independent places (§2.1.1). The remaining projections are $q, k = \text{L2Norm}(\text{Swish}(\text{ShortConv}(W_{q/k} x)))$, $v = \text{Swish}(\text{ShortConv}(W_v x))$, and a decay logit $z = W_\alpha^\uparrow W_\alpha^\downarrow x + b_\alpha^h$ from a low-rank projection plus a per-head bias.

By Equation (3), K3’s KDA layers have $\det A_t > 0$ at every token and head, and this would remain true at any R . That is a statement about what strict β forecloses, and it holds independently of our experiment. Whether a K3-style node would *in practice* benefit from $R > 1$ is an empirical question that our 1.4M-parameter probe cannot settle; we return to this in Section 7. We note the report contains no Householder, arity, or DELTAPRODUCT discussion, and no analysis of KDA’s state-tracking properties; the question we ask is not one K3 addresses.

A correction worth making explicitly. K3 introduces a *bounded* decay parameterization (§2.1.1, Eq. 5), $g = g_{\min} \cdot \text{Sigmoid}(e^{A_h} z)$ with $g_{\min} = -5$ fixed and A_h a learnable per-head log-scale initialized to 0, so $\alpha = \exp(g) \in (e^{g_{\min}}, 1)$. K3 reports this replaces the unbounded negative-softplus mapping $g = -e^{A_h} \text{Softplus}(z) \in (-\infty, 0)$ used by Kimi Linear [4]. It is tempting to read this floor as what forecloses reflections. It is not. Because $\alpha = \exp(g) > 0$ for any finite g , $\det \text{Diag}(\alpha) > 0$ under *both* parameterizations, so the floor cannot change the sign of $\det A_t$ at all. Determinant sign is governed solely by β . What forecloses odd permutations in K3 is the sigmoid-parameterized β of Equation (4), which Kimi Linear already had.

K3’s stated motivation for the floor is numerical, not representational. The reciprocal cumulative decay $1/\Gamma$ used to rescale keys within a chunk can grow without bound and overflow in finite precision; with $g_{\min} = -5$ the cumulative log-decay over a 16-token tile is bounded, so $1/\Gamma$ stays within BF16 range and *all* causal tiles can use dense Tensor Core matrix multiplications, eliminating the position-pair diagonal path (Fig. 3). No modelling or quality justification is offered, and the report contains no architecture ablations at all — so this is an absence of evidence, not evidence of absence. K3 does note the bounded form is related to lower-bounded recurrence gates in prior work; we read that as a relatedness remark appended after the numerical argument, since the paragraph’s topic sentence and the figure caption are both purely computational.

One consequence K3 does not discuss cuts against our own framing: the floor $\alpha > e^{-5} \approx 6.7 \times 10^{-3}$ is a genuine modelling constraint, since near-total single-step forgetting was representable under Kimi Linear’s unbounded range and is not under K3’s. Given that KDA is also K3’s only source of positional information, the decay range is load-bearing in a way the numerical framing does not convey. We raise it rather than let it be raised against us.

4 Experimental setup

Design. The experiment is a single design with four factors, run as 216 cells on one accelerator. The core is a 2×2 : arity $R \in \{1, 2\}$ crossed with β regime, over two tasks and twelve seed bundles. Three additional factors exist to remove the confounds a bare 2×2 cannot address, and each is crossed only where it is identified, because cells spent where a factor cannot move the answer are cells not spent on precision.

Capacity. Each $R=1$ arm has a parameter-matched twin (R1-P, R1-ref1-P) carrying the $R=2$ arms’ parameter count in feed-forward width rather than in extra factors. This makes arity separable from capacity *within* each regime. It is crossed with both regimes rather than only the strict one: the estimand is an interaction, so matching one contrast and not the other would leave the two halves differently constructed, and their difference would then mix the correction with the effect.

Learning rate. The two strict arms are additionally run at four peak rates spanning $33 \times$, $\{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$. This factor is crossed only with the strict arms and only over three bundles: the

reflection arms already sit at 97–100% on the affected metric, so there is no gap for a rate to close and sweeping them would buy nothing. This is the one deliberate asymmetry in the design and we flag it as a limitation rather than a feature.

Arity ladder. The strict regime is carried to $R \in \{3, 4\}$ (DP3-strict, DP4-strict), each with a capacity control matched to the rung above it (DP2-P3, DP3-P4). This factor is crossed only with strict β and only over the first six bundles. Its purpose is to reach $R \geq 3$, where the $k \leq 2$ bound of Grazi et al. [?] no longer holds, so a null there is a null against a theory rather than against an expectation. Section 5.6 explains why the rungs are matched to their neighbours rather than to a common $R=4$ target.

All ten arms (Table 1) use the same gated Householder mixer, the same Triton backend, 4000 steps, batch 64, and identical d_{model} , head count, state dimension and depth — so the recurrent state size, the variable every contrast must hold fixed, never varies. The β regime is a single scale on the sigmoid, $\beta = s \cdot \text{Sigmoid}(z)$ with $s \in \{1, 2\}$; it adds no parameters, and the two regimes have byte-identical parameter ledgers. The matched arms’ feed-forward width is solved analytically rather than searched — the cost is affine in the width — giving $d_{\text{ffn}} = 174$ and 1,399,756 non-embedding parameters against the $R=2$ arms’ 1,400,524, a -0.055% match, and the harness refuses to train if the realized match falls outside 0.5% . Post hoc, the strict arms reach $\beta_{\text{max}} = 1.0000$ and the reflection arms $\beta_{\text{max}} = 2.0000$, with no record violating its bound.

Block	Cells
2×2 core: $\{R1, \text{DP2-strict}, R1\text{-refl}, \text{Reflection}\} \times 2 \times 6$	48
Capacity: $\{R1\text{-P}, R1\text{-refl-P}\} \times 2 \times 6$	24
Learning rate: $\{R1, \text{DP2-strict}\} \times 4 \times 2 \times 3$	48
Total	120

Twelve of the learning-rate cells (those at the 10^{-3} level) are exact repeats of core cells — same arm, task, bundle, seed streams and rate — which we exploit as a replication check in Section 5.7.

Tasks. Two group word problems: given a sequence of generators of a finite group, predict the composed element. We use A_5 (order 60, non-solvable, *both* generators even) and S_5 (order 120, non-solvable, transposition generator *odd*). Chance accuracy is $1/60 = 1.67\%$ and $1/120 = 0.83\%$ respectively. The parity difference motivated the prediction Section 6 reports as falsified.

The metric, stated explicitly. Accuracy is the fraction of correct predictions averaged over *all* L positions of 64 evaluation sequences ($N = 64L$ scored tokens; we verified all 240 reported values are integer multiples of $1/64L$). Training words have length 3–40. Because the metric averages positions $1..L$, the nominal lengths are *nested*: $\text{acc}(512)$ includes positions ≤ 40 that are in-distribution. We therefore define *de-nested bands*

$$\text{band}(L_{i-1}, L_i) = \frac{L_i \text{acc}(L_i) - L_{i-1} \text{acc}(L_{i-1})}{L_i - L_{i-1}}, \quad (5)$$

the mean accuracy on positions in $(L_{i-1}, L_i]$ alone, and use these as the primary analysis. Only bands past position 40 are extrapolation.

Pairing, and why $n = 6$ helps. Six seed bundles, 1101–1106, each fixing four independent seed streams (initialization, data, task, evaluation), pairwise distinct in every record. Within a bundle every arm is scored on the same evaluation bank: we confirmed one distinct `eval_bank_sha256` per (task, bundle), shared by all six arms and by every level of the learning-rate factor, with twelve mutually distinct hashes across the twelve groups. The bank is derived from the evaluation seed alone, so this holds across the two execution batches as well — which is what makes the capacity factor a paired contrast against the core cells rather than merely a comparable one. Every contrast is a within-bundle difference cancelling evaluation-sampling noise identically in all arms. There were 0 train/eval collisions.

Statistics. For each cell we compute per-bundle differences and aggregate those. The interaction is differenced *per bundle before averaging*, $i_b = d_b^{\text{refl}} - d_b^{\text{strict}}$, preserving pairing; computing it from arm means with an unpaired standard error would be wrong. We report mean, $\text{se} = \text{sd}/\sqrt{6}$ (sd with $\text{ddof} = 1$), $t = \text{mean}/\text{se}$, and the one-sided 95% lower bound $L95 = \text{mean} - t_{0.95,5} \text{se}$, $t_{0.95,5} = 2.015$. We also report Holm-corrected outcomes across the band family, since we test many cells.

Compute and provenance. All 216 cells ran on one NVIDIA A10G apiece, 216/216 completing, for 16.28 GPU-hours (cells 228–317 s). No cell failed, 0 records were rejected, and 0 non-finite loss steps occurred anywhere. Cells were executed in four batches under four probe-source revisions — successively adding the capacity and learning-rate factors, the loss-summary instrumentation discussed in Section 5.5, the arity ladder, and the six additional bundles — and every record carries the revision that produced it. Section 5.7 uses the twelve duplicated cells to check that this does not bias the comparison; it does not.

Every cell’s evaluation bank is content-addressed by SHA-256, and the hash depends only on the (task, bundle) pair. Across all 216 records the 24 distinct (task, bundle) groups carry exactly 24 distinct hashes, so cells added in a later batch are paired with earlier ones rather than merely comparable to them. This is what licenses pooling the $n = 6$ and $n = 12$ analyses on the same bundles.

5 Results

5.1 Accuracy, and what the prefix metric hides

Table 2 gives both views. On the prefix-averaged metric the dominant feature is the β contrast: at $L = 128$ on A_5 , moving from strict to reflection β at fixed $R = 1$ takes accuracy from 25.74% to 82.83%, while doubling R at fixed strict β takes it from 25.74% to 27.75%.

Table 1: The six arms. “Non-emb. params” excludes both the token embedding and the output head; it is task-invariant, whereas total parameters differ by task because the head does. The β regime is parameter-free: the two regimes have byte-identical ledgers at equal R . The two -P arms carry the $R=2$ parameter count in feed-forward width instead of in a second Householder factor, which is what makes arity separable from capacity within a regime. $\bar{\beta}$ is the mean realized β pooled over both tasks and all six bundles.

Arm	β range	R	Non-emb. params	$\bar{\beta}$
R1	(0, 1)	1	998,092	0.899
R1-P	(0, 1)	1	1,399,756	0.878
DP2-strict	(0, 1)	2	1,400,524	0.779
DP2-P3	(0, 1)	2	1,802,188	0.803
DP3-strict	(0, 1)	3	1,802,956	0.744
DP3-P4	(0, 1)	3	2,204,620	0.755
DP4-strict	(0, 1)	4	2,205,388	0.725
R1-refl	(0, 2)	1	998,092	1.800
R1-refl-P	(0, 2)	1	1,399,756	1.767
Reflection	(0, 2)	2	1,400,524	1.658

De-nesting sharpens this and also disciplines it. In every extrapolation band, both strict arms sit within $2\times$ chance — R1 on A_5 scores 1.24–1.91% against 1.67% chance. The reflection arms are genuinely above chance for longer: Reflection on A_5 reaches 96.62% on positions 65–128 and 27.25% on 257–512. Table 3 makes the point quantitatively: a zero-parameter “correct below 40, at chance above” model reproduces both strict arms’ entire length curves to within 1.3pp, and fails on the reflection arms by 12–66pp. The honest reading is that *the strict- β arms do not extrapolate past the training range at all*, and the reflection arms partly do.

This also corrects an interpretation we would otherwise have offered. The strict-regime arity effect on the prefix metric decays monotonically with length (A_5 : +3.31, +3.50, +2.01, +0.98, +0.35pp), which invites a story about extra factors mattering less as the task hardens. It is dilution: the pure prediction $(40/L) \times \text{effect}(40)$ gives S_5 +2.67, +1.67, +0.83, +0.42, +0.21 against observed +2.67, +1.92, +0.82, +0.38, +0.21. That curve is one number divided by L , and we withdraw any interpretation of its shape.

5.2 The interaction

Table 4 gives the interaction under both metrics. On de-nested bands it is positive at uncorrected L95 in five of eight bands, from +22.14 to +57.09pp. De-nesting *strengthens* A_5 (+13.47 $\rightarrow$ +30.14 at 65–128; +35.28 $\rightarrow$ +57.09 at 129–256) and *eliminates* the two long-length S_5 cells, which reverse sign (+8.57 $\rightarrow$ -0.77; +4.44 $\rightarrow$ +0.31). Those two cells were significant only because prefix averaging carried the 41–64 signal forward; their per-bundle interaction values correlate at $\rho \geq 0.99$ with the $L = 64$ cell. We do not count them.

We must be equally direct about multiplicity. Across the ten prefix cells, *no* cell survives Holm or Bonferroni correction — Holm stops at the first comparison ($p = 0.00658$

against a threshold of 0.005). This is structural, not unlucky: with $df = 5$, clearing $\alpha = 0.005$ requires $|t| > 4.03$, and no prefix cell reaches it. On the eight de-nested bands, exactly one survives (S_5 , 41–64, $t = 4.26$, $p = 0.00401$) — and that one does not survive in the capacity-matched replication of Section 5.4, where it weakens to $t = 3.64$ and no band clears Holm. So the defensible statement is: *no* band-level interaction is established at family-wise $\alpha = 0.05$ in the better-controlled design, and several are positive and large but individually uncorrected. The count-based framing “ k of n cells significant” is especially weak here because the cells are not independent (mean within-task cross-length $r = 0.53$, several at 0.99–1.00).

5.3 β dominates arity, and it is free

Table 5 compares the two levers in the bands where both arms are measurable. On A_5 positions 41–64, widening β at fixed $R = 1$ buys +97.98pp ($t = 87.75$) against arity’s +3.81pp ($t = 2.40$), a ratio of $25.7\times$; on 65–128, +64.44 against +0.52, or $124.7\times$; on S_5 41–64, +30.64 against +0.67, or $45.5\times$. Beyond those bands the comparison stops being meaningful because all four core arms approach chance, and we report no ratio there. On the prefix metric the corresponding ratios run $6.3\times$ to $44.9\times$ across all ten cells, but we now regard those as contaminated and give them only for continuity.

The cost asymmetry is the part we consider most robust, because it does not depend on the effect size at all: the β widening adds *zero* parameters — both arms have 998,092 mixer parameters and byte-identical ledgers — whereas doubling arity costs +40.3% of mixer parameters (+39.7% / +39.1% of total, by task). Whatever the true effect sizes, one intervention is free and the other is not.

We do not claim arity does nothing. An earlier framing of this work said arity’s effect was “not distinguishable from zero”. That is true cell-by-cell and misleading overall. The strict-regime arity effect is positive in 10 of 10 prefix cells (one-sided sign test $p = 0.00098$, more significant than any single interaction cell we report), and pooling all 60 per-bundle differences gives +1.61pp, $se = 0.61$, $t = 2.66$ — though those 60 values are nested and not independent, so the pooled test overstates its own precision. Our design also had little power: the minimum detectable effect at 80% power is 7.4–12.5pp at short lengths, so a real +3.3pp arity effect would have been invisible to us. The accurate statement is that *arity’s benefit under strict β is consistently positive, one to two orders of magnitude smaller than the β effect, and below this design’s per-cell resolution* — not that it is absent.

5.4 The capacity factor: arity separated from parameter count

The $R=1$ arms of the core square have 998,092 non-embedding parameters and the $R=2$ arms 1,400,524, +40.3%. A raw “ $R=2$ beats $R=1$ ” comparison therefore confounds arity with capacity, which is why the capacity factor exists.

Table 2: Mean accuracy (%), $n = 6$ bundles. **Left:** the prefix-averaged metric our harness reports, over positions 1.. L . **Right:** the same data de-nested into position bands via Equation (5), which is what actually measures behaviour at new lengths. Chance is 1.67% (A_5) and 0.83% (S_5); *italic* marks band values within $2\times$ chance. Both strict arms are at chance in every extrapolation band — the prefix metric disguises this as graceful decay.

Arm	Task	Prefix-averaged, positions 1.. L					De-nested bands, positions in $(L_{i-1}, L_i]$				
		40	64	128	256	512	1–40	41–64	65–128	129–256	257–512
R1	A_5	79.19	49.96	25.74	13.83	7.82	79.19	<i>1.24</i>	<i>1.53</i>	<i>1.91</i>	<i>1.81</i>
DP2-strict	A_5	82.49	53.45	27.75	14.81	8.17	82.49	5.05	<i>2.04</i>	<i>1.87</i>	<i>1.53</i>
R1-refl	A_5	99.99	99.70	82.83	46.47	23.63	99.99	99.22	65.97	10.11	<i>0.78</i>
Reflection	A_5	100.00	100.00	98.31	82.73	54.99	100.00	100.00	96.62	67.15	27.25
R1	S_5	74.06	46.98	23.65	12.51	6.63	74.06	<i>1.86</i>	<i>0.31</i>	<i>1.37</i>	<i>0.76</i>
DP2-strict	S_5	76.73	48.90	24.47	12.89	6.84	76.73	<i>2.53</i>	<i>0.04</i>	<i>1.31</i>	<i>0.79</i>
R1-refl	S_5	97.07	72.86	37.30	19.58	9.99	97.07	32.50	<i>1.75</i>	<i>1.85</i>	<i>0.41</i>
Reflection	S_5	98.09	88.45	56.04	28.53	14.64	98.09	72.40	23.62	<i>1.02</i>	<i>0.75</i>

Table 3: Why we de-nest. Maximum absolute error, over $L \in \{64, 128, 256, 512\}$, of the *zero-free-parameter* prediction $\text{acc}(L) = \frac{40}{L} \text{acc}(40) + \frac{L-40}{L} \cdot \text{chance}$, i.e. “correct below 40, at chance above”. The strict arms are fit to within 1.3pp: their apparent length curve carries no extrapolation signal. The reflection arms deviate by 12–66pp, so they genuinely do extrapolate.

Arm	A_5 max err	S_5 max err
R1	0.16	0.38
DP2-strict	1.27	0.64
R1-refl	50.44	11.87
Reflection	65.92	26.84

It is worth being explicit about an argument we do *not* make. One could try to bound the confound by observing that the strict-regime arity effect is small ($\leq +3.50\text{pp}$) and reading that as a ceiling on what 402,432 parameters buy. That reasoning is circular here: as Table 2 shows, both strict arms are at chance in every extrapolation band, and a capacity contrast measured between two floored models reads ≈ 0 regardless of what capacity is worth. The bound is uninformative rather than reassuring. The factor settles by measurement what the bound cannot settle by argument.

The interaction is not a capacity artifact. Table 6 gives the same de-nested contrast computed with the unmatched and the matched $R=1$ cells. On A_5 at positions 129–256 the interaction is +58.88pp matched against +57.09pp unmatched (se 14.85, $t = 3.96$, L95 +28.95); at 65–128 it is +51.43 against +30.14. Matching moves the A_5 estimates *up*, not down. The strict-regime arity effect stays ≈ 0 in every extrapolation band under matching (−0.23 to +3.10pp) — and this version of that statement is informative, because R1-P is no longer capacity-starved relative to DP2-strict.

At $n = 6$ matching cost us our one significance claim; at $n = 12$ it does not. We report both, because the first was the state of this paper for a week and the reason for the second is arithmetic rather than insight. Under Holm at $\alpha = 0.05$,

the unmatched cells at $n = 6$ yield exactly one survivor (S_5 , 41–64, $t = 4.26$, $p = 0.00401$); matched, that band weakens to $t = 3.64$ and *no* band survives, the strongest being A_5 129–256 at $p = 0.00535$ against a first-step threshold of 0.00500.

Six further seed bundles (1107–1112, identical protocol, same frozen evaluation banks) resolve it. The standard error of a paired mean falls as $1/\sqrt{n}$, and nothing about the point estimates needed to change for the family to clear: at $n = 12$, **four** of the eight matched extrapolation bands survive one-sided Holm (Table 7). The strongest is S_5 41–64 at $t = 6.34$, $p = 0.00003$ — the same band that missed at $n = 6$. A_5 129–256, the band that missed by 0.00035, now reads $p = 0.00282$ against a threshold of 0.00833.

This is worth stating precisely, because “more seeds made it significant” is also what a fishing expedition sounds like. The band family, the estimand, the correction and the one-sided convention were all fixed before these bundles ran; the only quantity that changed is n . What $n = 6$ established was that the effects were large and the evidence was too thin to survive a family-wise correction over eight bands. What $n = 12$ establishes is that four of them do. The point estimates at $n = 12$ are, if anything, slightly smaller than at $n = 6$ (A_5 129–256: +40.94pp against +58.88pp), which is the direction regression to the mean predicts and the opposite of what selective reporting would produce.

5.5 The learning-rate factor: a generalization gap, not an optimization failure

The remaining alternative to an expressivity account is an optimization one, and the design addresses it in two steps: first by characterizing what the strict arms actually do, then by sweeping the most obvious optimization lever.

What the loss traces do and do not show. Ten of the 48 core cells end with a final logged loss above 0.1, and *all ten* are strict- β (R1 6/12, DP2-strict 4/12; both reflection arms 0/12). A perfectly asymmetric stability failure between the arms being compared is the signature of an optimization difference, not only an expressivity one. Within the 24 strict

Table 4: The $\beta \times R$ interaction, differenced per bundle before averaging. **Left:** on the prefix-averaged metric. **Right:** on de-nested position bands, our primary analysis. L95 is the one-sided 95% lower bound ($t_{0.95,5} = 2.015$); **bold** marks L95 > 0 uncorrected. Seven of ten prefix cells and five of eight bands are positive at uncorrected L95. Under Holm correction across the family, **zero** of the ten prefix cells and **one** of the eight bands survive (marked †): with $df = 5$, $p < 0.00625$ needs $|t| > 4.03$ and our maximum band t is 4.26. This table uses the unmatched $R=1$ cells at $n = 6$; the capacity-matched square at $n = 12$ yields **four** Holm survivors (Section 5.4, Table 6). Note S_5 at 129–256 and 257–512 *reverse sign* once de-nested — they were significant only by inheriting the 41–64 signal through prefix averaging.

Prefix-averaged, positions 1..L					De-nested bands				
Task, L	Inter.	se	t	L95	Task, band	Inter.	se	t	L95
$A_5, 40$	−3.30	4.25	−0.78	−11.87	$A_5, 41\text{--}64$	−3.03	1.53	−1.97	−6.12
$A_5, 64$	−3.20	3.09	−1.03	−9.43	$A_5, 65\text{--}128$	+30.14	9.34	3.23	+11.32
$A_5, 128$	+13.47	4.74	2.84	+3.93	$A_5, 129\text{--}256$	+57.09	16.08	3.55	+24.69
$A_5, 256$	+35.28	9.38	3.76	+16.37	$A_5, 257\text{--}512$	+26.74	12.56	2.13	+1.43
$A_5, 512$	+31.01	10.70	2.90	+9.45	$S_5, 41\text{--}64$	+39.23†	9.21	4.26	+20.66
$S_5, 40$	−1.65	2.42	−0.68	−6.52	$S_5, 65\text{--}128$	+22.14	8.80	2.52	+4.40
$S_5, 64$	+13.68	4.23	3.23	+5.15	$S_5, 129\text{--}256$	−0.77	0.79	−0.98	−2.35
$S_5, 128$	+17.91	6.33	2.83	+5.15	$S_5, 257\text{--}512$	+0.31	0.24	1.31	−0.17
$S_5, 256$	+8.57	3.19	2.69	+2.15					
$S_5, 512$	+4.44	1.68	2.64	+1.05					

cells, log final loss correlates with accuracy at $r = -0.47$ ($L = 40$).

It would be wrong to call those ten runs unconverged, and an earlier draft of this paper did. The trace logs one minibatch every 500 steps, so its final entry is a *single batch drawn at step 3999*, where the one-cycle schedule has decayed the rate to 4×10^{-9} and the weights are effectively frozen. All ten reach 10^{-4} or below after warmup; R1 on S_5 bundle 1106 reads 10^{-4} at step 3000 and 0.82 at step 3999. Those models fit the training set. What the traces do show is *instability*: two runs are above 0.05 at five of seven sampled steps, which is sustained oscillation rather than one unlucky draw. Instability is the correct name for the phenomenon, and it remains perfectly asymmetric across the regimes.

The gap is in-distribution. Naming it correctly sharpens it. The strict arms fit to ≈ 0 training loss and still score 68–89% at $L = 40$ — a length *inside* the training range 3–40 — where R1-ref1 scores 99.99% (A_5) and 97.07% (S_5). That is a generalization gap on data drawn from the training distribution. It is not length extrapolation, and the nesting artifact of Section 5.1 does not touch it.

No learning rate closes it. Table 8 gives the swept factor. Across $33\times$ in peak rate, *no* setting closes the gap. The best strict cell anywhere is DP2-strict on A_5 at 91.33%, still 8.67pp short of the reflection ceiling; on S_5 the best is 84.02%, 14.06pp short. At the two highest rates the strict arms are not merely fitting but fitting *stably* — tail-median training loss 0.0006–0.0084 over the last 200 steps — and still cap at 86.8%. They fit the training set and do not generalize, and no rate repairs that by fitting harder.

Two caveats keep this from being conclusive. On three of the four arm/task pairs the optimum is interior to the grid, so the grid brackets it; but DP2-strict on A_5 peaks at the top

rate we tried, so we have not bracketed its optimum and cannot exclude a still-better rate above 10^{-2} . And the learning rate is not the only optimization lever: schedule shape, weight decay, gradient clipping, and initialization scale are untested. What we can say is that the single most likely optimization explanation, tested over a $33\times$ range, does not account for a 9–21pp deficit. We do continue to flag that $\beta = 2$ doubles the effective write-gate gradient scale, so the reflection arms’ perfect stability record may itself be an optimization-geometry effect.

5.6 The arity ladder: $R = 3$ and $R = 4$ under strict β

Everything above compares $R=1$ against $R=2$, which is exactly the range in which Grazi et al.’s Proposition 1 [?] already tells us what to expect. The interesting question is whether arity pays where that bound stops applying. We therefore extend the strict regime to $R \in \{3, 4\}$, with a capacity control at each rung.

Adjacent rungs, not a common target. The obvious design — match everything to the $R=4$ parameter count — is the wrong one here, and the reason is quantitative. Parameters are exactly affine in R at this geometry: the mixer costs $997,068 + 402,432(R - 1)$. Matching an $R=1$ arm up to $R=4$ therefore requires a feed-forward block of 1,207,296 parameters, which is $1.21\times$ that arm’s own mixer and 55% of the resulting model. That is not a control but a different architecture, and we can measure what it would cost: adding merely one third of that dose ($\text{ffn_dim} = 174$) to R1-ref1 costs −16.35pp at A_5 41–64 and −20.54pp at 65–128 — the two bands the headline interaction lives in. A control damaged in the bands that carry the result biases the comparison toward “arity pays” for a reason unrelated to arity.

We instead compare *adjacent* rungs, each matched by the same validated $\text{ffn_dim} = 174$ increment the existing con-

trols carry: DP2-P3 ($R=2$ grown to $R=3$'s count, -0.043%) and DP3-P4 ($R=3$ grown to $R=4$'s, -0.035%). Matching only runs upward: `solve_ffn_dim` refuses a target below the feed-forward-free cost.

One band moves, and it is not an extrapolation band. On A_5 at positions 41–64, strict- β accuracy climbs monotonically with R : 1.24% ($R=1$), 5.05% ($R=2$), 16.79% ($R=3$), 19.90% ($R=4$), against a 1.67% chance floor. This is the one place in the entire study where strict β does something, and it contradicts what our own $R \leq 2$ data predicted. Every other cell — both tasks, all three longer bands, all four arities — sits within noise of chance.

The matched effect is mostly damage to the control. Reporting the matched contrast alone would overstate this badly, so we decompose it. Writing the matched effect as the raw arity effect plus whatever the control's feed-forward block does to the control itself, at A_5 41–64:

Rung	Raw R	Control harm	Matched
$R=3$ vs $R=2$	+11.74	+2.73	+14.47
$R=4$ vs $R=3$	+3.11	+13.56	+16.68

At $R=3$ the matched estimate is mostly a real arity effect (+11.74pp, $t = 2.62$ unmatched). At $R=4$ it is mostly not: the raw effect is +3.11pp at $t = 1.06$, and 81% of the matched +16.68pp is the DP3-P4 control losing 13.56pp to its own feed-forward block. The honest reading is that $R=3$ buys something at short range and $R=4$ buys little beyond it.

What this does and does not establish. Neither ladder contrast survives Holm over the 16-contrast family it belongs to: at $df = 5$, $t = 2.60$ is $p \approx 0.024$ against a first-step threshold of 0.003. Both live in a single task and a single band, at $n = 6$. We report the ladder as a signal worth following, not a result — and note that it does not disturb the paper's comparison, since the β effects in Table 7 are +22 to +52pp where arity's best is +15pp, and β remains free while $R=4$ costs +121% in parameters.

5.7 A replication check the design gives for free

Twelve cells appear twice: the 10^{-3} level of the learning-rate factor repeats twelve core cells at identical arm, task, bundle, seed streams and rate. The two instances were built from different image revisions, so the pair measures end-to-end run-to-run reproducibility rather than seed variance.

Agreement is unbiased but not tight. Mean signed difference in accuracy is -0.31 pp at $L = 40$ and within ± 0.4 pp at every other length, so there is no systematic drift between revisions. The scatter is larger: sd 3.63pp at $L = 40$, falling to 0.37pp at $L = 512$, with a worst single cell of 9.18pp (DP2-strict, A_5 , bundle 1103). We report this for two reasons. It justifies the paired design — a within-bundle contrast differences this noise away, an unpaired comparison of single cells does not — and it sets a floor on how finely any single cell of this experiment should be read. Differences below

Table 5: β range versus arity, on de-nested bands, restricted to the three extrapolation bands where the compared arms are above chance. Both interventions start from the same $R=1$ strict baseline; Δ is a paired mean difference in points. The β intervention is parameter-free; arity costs +40.3% mixer parameters. Bands past these are omitted because all arms approach chance, making the ratio meaningless.

Task, band	Lever	Δ	t	Ratio
A_5 , 41–64	β	+97.98	87.75	
A_5 , 41–64	arity	+3.81	2.40	25.7×
A_5 , 65–128	β	+64.44	7.55	
A_5 , 65–128	arity	+0.52	1.36	124.7×
S_5 , 41–64	β	+30.64	6.05	
S_5 , 41–64	arity	+0.67	1.02	45.5×

Table 6: The capacity factor. The same de-nested interaction computed with the unmatched $R=1$ cells ($R1/R1$ -ref1, 998,092 non-embedding parameters) and with their matched twins ($R1$ -P/ $R1$ -ref1-P, feed-forward-widened to 1,399,756 against the $R=2$ arms' 1,400,524, -0.055%). Both use the same six bundles and byte-identical evaluation banks, so the columns are paired. Matching *raises* the A_5 mid-range estimates and leaves the strict-arity effect at ≈ 0 . Both columns are $n = 6$; Table 7 carries the matched square to $n = 12$, where four bands clear Holm. **Bold** marks uncorrected $L95 > 0$.

Task	Band	Unmatch.	Capacity-matched			
		Inter.	Inter.	se	t	L95
A_5	1–40	−3.30	−2.46	5.90	−0.42	−14.35
A_5	41–64	−3.03	+14.03	11.54	1.22	−9.22
A_5	65–128	+30.14	+51.43	14.77	3.48	+21.67
A_5	129–256	+57.09	+58.88	14.85	3.96	+28.95
A_5	257–512	+26.74	+25.42	12.53	2.03	+0.18
S_5	1–40	−1.65	+5.28	3.50	1.51	−1.78
S_5	41–64	+39.23	+52.52	14.43	3.64	+23.44
S_5	65–128	+22.14	+21.77	9.08	2.40	+3.47
S_5	129–256	−0.77	+0.26	1.03	0.25	−1.81
S_5	257–512	+0.31	−0.30	0.35	−0.87	−1.01

roughly 4pp at short lengths, between individual cells, are not resolvable by this apparatus.

6 Two failed explanations

We stated a mechanistic prediction before running the sweep. It failed. The replacement we then offered also fails against the same data. We report both.

The parity prediction, and its falsification. Equation (3) says reflection β unlocks negative determinants, i.e. odd permutations. Our tasks differ exactly in whether odd permutations are demanded: S_5 's transposition generator is odd, so composing a word requires elements of determinant -1 ; in A_5 both generators are even. We predicted a *large* interaction on S_5 and a *small* one on A_5 . We observe the opposite in the bands that matter: de-nested, the interaction is +30.14 and +57.09pp on A_5 at 65–128 and 129–256, against -0.77

Table 7: The capacity-matched interaction at $n = 12$: the six original bundles plus 1107–1112, identical protocol and the same frozen evaluation banks. p is one-sided; † marks the four bands that survive Holm at $\alpha = 0.05$ over the eight extrapolation bands (1–40 is inside the training range and is excluded from the family, reported for reference). **Bold** marks $L95 > 0$ uncorrected, with $t_{0.95,11} = 1.796$. Nothing about the design changed between $n = 6$ and $n = 12$ except the number of bundles; the point estimates are slightly *smaller* here than in Table 6, and four bands clear anyway.

Task	Band	Inter.	se	t	L95	p
A_5	1–40	−1.49	2.97	−0.50	−6.83	0.687
A_5	41–64	+7.47	5.86	1.27	−3.06	0.115
A_5	65–128	+32.08	10.59	3.03	+13.06	0.0057†
A_5	129–256	+40.94	11.94	3.43	+19.49	0.0028†
A_5	257–512	+20.86	9.00	2.32	+4.68	0.020
S_5	1–40	+4.52	2.24	2.02	+0.50	0.034
S_5	41–64	+51.67	8.15	6.34	+37.04	0.00003†
S_5	65–128	+22.57	5.94	3.80	+11.90	0.0015†
S_5	129–256	+0.21	0.51	0.40	−0.71	0.347
S_5	257–512	+0.14	0.26	0.53	−0.33	0.302

and +0.31pp on S_5 in the same bands. Only the 41–64 band favours S_5 . The reachability-of-reflections story does not explain our result. As Section 2.3 notes, it never should have: Equation (3) predicts no interaction at $R = 2$ in the first place.

The headroom explanation, also contradicted. We then proposed that $\beta > 1$ helps by making each write more expressive regardless of parity, with A_5 (order 60) simply easier than S_5 (order 120) so its reflection arms retain headroom where S_5 ’s have collapsed. This predicts the interaction should track available headroom. It does not. Across the ten prefix cells, the correlation between interaction and headroom ($100 - \text{acc of R1-refl}$) is only $r = +0.53$; within A_5 it fits well ($r = +0.93$) but within S_5 there is no relationship ($r = +0.28$, $\rho = +0.10$), and using the Reflection arm’s de-nested band accuracy as the regressor gives the *wrong sign* ($r = -0.33$). The specific claim we had drafted — that the S_5 reflection arms at 19.58% “have little room left” — is also indefensible: 19.58% against a 0.83% chance floor is ~ 80 pp of headroom, more than A_5 at $L = 256$ (46.47%), where the interaction is four times larger.

Where that leaves the mechanism. We have a measured interaction, a falsified pre-registered mechanism, a falsified fallback, and a determinant argument that predicts no interaction at $R = 2$. We therefore offer *no* mechanism. A reader should treat the interaction as an unexplained empirical regularity — one that survives the two controls the design carries (Sections 5.4–5.5) but has no account of *why*. Distinguishing the remaining candidates needs a solvable-group control and a difficulty-matched non-solvable pair, neither of which we ran. The third item on that list — an $R = 3$ arm, where Equation (3) does predict something because the $k \leq 2$ hypothesis of Grazi Prop. 1.3 fails — we did run, and Section 5.6 re-

Table 8: No learning rate closes the in-distribution gap. Accuracy at $L = 40$, a length *inside* the training range 3–40, for the strict arms at four peak rates ($n = 3$ bundles each), against the reflection arms at the design’s default rate as the ceiling. “Fit” is tail-mean training loss over the last 200 steps: at the two highest rates the strict arms fit stably and still fall 9–21pp short, so the deficit is not something a better rate repairs by fitting harder. The optimum is interior to the grid for three of four arm/task pairs; DP2-strict on A_5 peaks at the top rate, so its optimum is not bracketed.

Task	Arm	Peak LR	Acc. $L=40$	Fit
A_5	R1	3×10^{-4}	56.78	0.538
A_5	R1	10^{-3}	76.12	0.163
A_5	R1	3×10^{-3}	83.66	0.093
A_5	R1	10^{-2}	83.61	0.082
A_5	DP2-strict	3×10^{-4}	62.34	0.419
A_5	DP2-strict	10^{-3}	86.63	0.063
A_5	DP2-strict	3×10^{-3}	88.83	0.049
A_5	DP2-strict	10^{-2}	91.33	0.031
A_5 ceiling (R1-refl/Reflection)			99.99/100.00	
S_5	R1	3×10^{-4}	55.44	0.604
S_5	R1	10^{-3}	71.07	0.251
S_5	R1	3×10^{-3}	76.73	0.170
S_5	R1	10^{-2}	71.97	0.246
S_5	DP2-strict	3×10^{-4}	59.06	0.524
S_5	DP2-strict	10^{-3}	76.13	0.190
S_5	DP2-strict	3×10^{-3}	84.02	0.101
S_5	DP2-strict	10^{-2}	82.28	0.115
S_5 ceiling (R1-refl/Reflection)			97.07/98.09	

ports it. It does not rescue the mechanism: $R=3$ helps only on A_5 at 41–64, a band *below* the training cutoff’s reach where a determinant argument predicts nothing in particular, and it leaves every extrapolation band at chance.

7 Limitations

The two confounds a reader would raise first — capacity and optimization — are factors in the design and are reported in Sections 5.4–5.5. What follows is what the design does *not* address.

7.1 Sample size, multiplicity, and influential bundles

The core square is $n = 12$; every other factor — learning rate, the arity ladder, and the leave-one-out diagnostics below — remains at $n = 6$, so the precision claims differ by factor and we state which applies where.

At $n = 6$ paired contrasts are informative but imprecise: interaction standard errors run 1.68–16.08pp, and the Reflection arm’s between-bundle spread on A_5 is large (sd 20.19pp at $L = 256$). At that size no prefix cell and only one band survived Holm correction, and with $df = 5$ none could have cleared Bonferroni even in principle. Doubling to $n = 12$ on the matched square is what changes this: four bands clear Holm (Table 7), and $df = 11$ makes Bonferroni reachable — S_5 41–64 at $p = 0.00003$ clears $0.05/8 = 0.00625$ by two or-

ders of magnitude. *The arity ladder has no such backing*: it is $n = 6$, and neither of its two positive contrasts survives correction over its own family. Leave-one-bundle-out shows the two long-length S_5 prefix cells lose significance if either of two bundles is dropped — consistent with their sign reversal under de-nesting. The A_5 result is more robust: dropping the most extreme bundle (1103) takes $L = 256$ from $+35.28$ to $+29.18$ (L95 $+10.56$) and $L = 512$ from $+31.01$ to $+22.26$ (L95 $+6.18$), both still positive. We report L95 values, not point estimates, as the defensible claim.

7.2 Ceiling and floor censoring

Of 240 reported values, 26 are exactly 100.00% and 35 exceed 99%; of the de-nested band values, 107 of 240 lie within $2\times$ chance and 10 band-cells fall strictly below it. At $L = 40$ the reflection arms score 99.99% and 100.00% on A_5 , the latter with zero variance across all six bundles, so the $L = 40$ interaction is mechanically bounded and its negative sign (-3.30pp) is not interpretable. As a robustness check we recomputed the interaction on Haldane–Anscombe-corrected log-odds: sign and significance broadly survive (9 of 10 prefix cells positive at uncorrected L95), but the largest interaction moves from $L = 256$ to $L = 128$ on A_5 , and S_5 at $L = 40$ becomes significantly *positive*. So the raw-scale magnitudes are partly an artifact of pp differences being maximized near 50%, and our reading of the $L = 40$ cells as uninformative is right on A_5 but wrong on S_5 .

7.3 Scale, scope, and the K3 claim

Our models have 1.0–1.4M mixer parameters, no feed-forward blocks, and train for 4000 steps on synthetic group word problems. Kimi K3 has 2.78T total parameters. **Nothing in this paper is direct evidence about the behaviour of a frontier-scale model.** We separate the two K3 statements accordingly. The determinant observation — that strict β forecloses negative-determinant transitions at every arity — follows from Equation (3) alone, is regime-agnostic, and does not depend on this experiment; we consider it sound. The empirical prediction that K3-style nodes would not *benefit* from $R > 1$ does depend on this experiment, and given the caveats above we present it as a hypothesis motivating a scaled replication, not as a finding. Group word problems are also a deliberately adversarial probe chosen because they discriminate; language modelling may weight these mechanisms differently.

We also flag a tension with the prior work we rely on. DELTAPRODUCT reports that its models “fail to learn even the training context-length when restricted to the standard eigenvalue range $[0, 1]$, regardless of the number of Householder transformations n_h ”. Our strict- β arms do learn the training range (74–82% at $L = 40$ at our default rate, and up to 91.3% once the rate is tuned, Table 8), while reaching near-zero training loss throughout. So the disagreement is not that their models fail to fit and ours fit: ours fit and then generalize imperfectly. Either the tasks differ materially or our strict arms are not the regime they describe; we cannot say which.

7.4 Other limitations

We measure $R \in \{1, 2\}$ only, so we cannot speak to $R \geq 3$ — where, as noted, the theory actually predicts something different. All arms share one mixer implementation and Triton backend, so a kernel bug interacting with the β scale would not be caught by our design. The main 2×2 sweep uses a single learning-rate schedule per arm; Section 5.5 sweeps the rate for the strict arms only, on three bundles rather than six, so the reflection arms remain untuned and we cannot rule out that a different rate would move *them* as well — though they are already at 97–100% on the metric in question, so there is little room for it to matter.

8 Related work

Delta-rule linear attention. The delta rule as a fast-weight update goes back to Schlag et al. [6]; DeltaNet [8] made it trainable at scale with a chunkwise-parallel form, and gated DeltaNet [9] added the channel-wise forget gate that makes Equation (1) the modern default. Mamba-2 [1] supplies the structured-state-space view of the same gating. Kimi Linear [4] introduced KDA, and Kimi K3 [5] deploys it as the majority attention layer at frontier scale.

Arity and negative eigenvalues are one research programme. The two lines of work our paper sits between share most of their authors. Grazi et al. [3] established that admitting negative eigenvalues unlocks state tracking otherwise out of reach — the result our Section 2.3 specializes and which subsumes our determinant observation. DELTAPRODUCT [7] then raised arity, and its §5.2 reports the finding that most directly prefigures ours: “Unexpectedly, S_4 and A_5 can extrapolate robustly using only $n_h = 2$ despite the theorem suggesting 3 and 4, respectively.” Two details matter for us. They attribute the efficiency to S_4 and A_5 being isomorphic to subgroups of $SO(3, \mathbb{R})$, and they observe a head that learns $\beta = 2$ on *both* factors — two true reflections composing to a rotation. And their §5.2 uses the extended eigenvalue range throughout, reporting the failure under $[0, 1]$ quoted in Section 7.3. That sentence is the closest prior statement to our result. Our contribution is to convert it from a training-time observation reported in passing into a paired factorial estimate at length extrapolation — with the caveats above. Our result is a refinement of their programme, not a challenge to it.

The operator is not new, and we are precise about what is prior. The per-channel-gated delta-rule operator we measure is computed by public kernels for generalized (diagonal-plus-low-rank) delta-rule updates, and is established at scale: EDA [2] trains dense 2.5B and MoE 25B-A2.8B models on it — though it appears there as the *baseline*, explicitly credited to KDA, EDA’s own contribution being a second, independently addressed erase factor. To avoid overclaiming on DELTAPRODUCT’s behalf: its Appendix B.4 Eq. 6 does place a per-channel $\text{diag}(w)$ inside a product at arbitrary n_h , but the factors are asymmetric RWKV-7 matrices rather than Householders, the construction is offered explicitly as buying ex-

pressivity “at the cost of stability”, and it is used for a theory result with no experiments; DELTAPRODUCT’s own gated variant uses a *scalar* gate outside the product. We contribute no operator, no kernel and no theorem — only a controlled measurement.

9 Conclusion

In a gated delta-rule node, the write-strength range β and the Householder arity R are not interchangeable routes to expressivity. Across 216 paired cells on two non-solvable group word problems, widening β produced large gains in every position band where the compared arms were above chance — $25.7\times$ to $124.7\times$ the gain from doubling arity — at zero parameter cost, against arity’s $+40.3\%$. The $\beta \times R$ interaction was positive in five of eight de-nested bands. The determinant argument of Section 2.3, which is not ours, establishes independently of any experiment that a strict- β node such as Kimi K3’s KDA layer cannot obtain a negative-determinant transition at any arity.

The two alternative explanations the design was built to exclude are excluded. The effect is not capacity: with $R=1$ controls widened to within 0.055% of the $R=2$ arms in *both* regimes, the interaction is intact, and at $n=12$ four bands survive one-sided Holm correction, the strongest at $p=0.00003$. It is not the learning rate: across $33\times$ in peak rate, no setting closes the strict arms’ deficit at a length *inside* the training range, where they reach stable near-zero training loss and still fall 9–21pp short. That last result also changes what the deficit *is* — a generalization gap on in-distribution data, not a training failure.

What remains open we state plainly. We have no mechanism: our pre-registered explanation is falsified and so is the fallback. The arity ladder we added to test the one place a determinant argument still had room — $R \geq 3$, where Grazi’s $k \leq 2$ hypothesis fails — does not rescue it: $R=3$ helps on a single task in a single short band, $R=4$ ’s apparent matched gain is 81% damage to its own control, and every extrapolation band stays at chance. One arm of the rate factor is unbracketed at the top of the grid, optimization levers other than the rate are untested, and our own replication check says single cells of this apparatus are not resolvable below roughly 4pp at short lengths. The remaining fixes are specified and cheap: a solvable-group control, a difficulty-matched non-solvable pair, and a wider optimization search. What we can defend is narrow but it is now well controlled: one lever is free and appears to matter a great deal, the other is expensive and appears to matter little, the difference is explained by neither capacity nor learning rate, and we do not yet know why or whether it survives at scale.

References

- [1] T. Dao and A. Gu. Transformers are SSMS: Generalized models and efficient algorithms through structured state space duality. *ICML*, 2024. arXiv:2405.21060.
- [2] X. Li, C. Zhang, H. Luo, X. Lin, Z. Wang, Z. Qiu, Y. Mao, L. Chen, M. Yuan, M. Sun, H. Jiang, S. Zhang, R. Men, W. Hu, G. Cheng, B. Zheng, D. Liu, and J. Zhou. Erase-then-Delta attention: Decoupling erase and write addresses in delta-rule linear attention. arXiv:2606.26560, 2026.
- [3] R. Grazi, J. Siems, A. Zela, J. K. H. Franke, F. Hutter, and M. Pontil. Unlocking state-tracking in linear RNNs through negative eigenvalues. *ICLR*, 2025. arXiv:2411.12537.
- [4] Kimi Team. Kimi Linear: An expressive, efficient attention architecture. arXiv:2510.26692, 2025.
- [5] Kimi Team. Kimi K3: Open frontier intelligence. arXiv:2607.24653, 2026.
- [6] I. Schlag, K. Irie, and J. Schmidhuber. Linear transformers are secretly fast weight programmers. *ICML*, 2021. arXiv:2102.11174.
- [7] J. Siems, T. Carstensen, A. Zela, F. Hutter, M. Pontil, and R. Grazi. DeltaProduct: Improving state-tracking in linear RNNs via Householder products. *NeurIPS*, 2025. arXiv:2502.10297.
- [8] S. Yang, B. Wang, Y. Zhang, Y. Shen, and Y. Kim. Parallelizing linear transformers with the delta rule over sequence length. *NeurIPS*, 2024. arXiv:2406.06484.
- [9] S. Yang, J. Kautz, and A. Hatamizadeh. Gated delta networks: Improving Mamba2 with delta rule. *ICLR*, 2025. arXiv:2412.06464.